

MCS-023 (DBMS)

1. How do weak entities differ from strong entities ? Discuss with the help of an example. Briefly discuss the role of keys in identification of the entity type, i.e. weak or strong.
2. BCNF is stronger than 3NF'. Give suitable example in support of your justification.
3. Draw ANSI-SPARC 3-level architecture of DBMS. Write the languages associated with different levels of ANSI-SPARC 3-level architecture.
4. In terms of DBMS, what is a transaction ? Discuss ACID properties of transaction.
5. Explain database recovery using a system log with the help of an example
6. Can we use Binary Search Tree (BST) for indexing? Justify.
7. Differentiate between serial schedule and serializable schedule.

Functional Dependency & Normalization

8. What do you understand by the term “functional dependency” ? Explain the following functional dependencies, with suitable example for each :
 - (i) Full functional dependency
 - (ii) Partial functional dependency
 - (iii) Transitive functional dependency
 - (iv) Trivial functional dependency
9. What do you understand by the term ‘Normalization’ in DBMS? Write statement for second normal form (2NF), and discuss the insert, delete and update anomalies associated with 2NF.

Concurrency Control & Deadlock

10. Explain the Two-Phase Locking Protocol. How does it guarantee serializability?
11. What is deadlock and how can it be prevented? Describe common deadlock-prevention schemes.

ER Modeling

12. Draw an ER diagram for an open university system covering all the functionalities and also derive corresponding relational schema.
13. Draw an ER diagram for an academic institution... Transform your ER diagram into a relational database.

Indexing Structures

14. Compare primary, secondary and clustering indexes. Which of these indexes are dense and which are not? How is implementation of clustering indexes performed?
15. How do B-tree indexes differ from binary search tree indexes?
16. What is a B⁺-tree? Why is a B⁺-tree better structure than a B-Tree for implementation of an index sequential file?

SQL & Views

17. Write SQL commands for each of the following:
 - (i) Creation of views;
 - (ii) Creation of sequences;
 - (iii) Outer join;
 - (iv) To give access permission to any user.

Data Replication & Fragmentation

18. Why is data replication useful in Distributed Database Management System (DDBMS)? Briefly discuss the terms ‘Complete Replication’ and ‘Selective Replication’.
19. Explain the term data replication and data fragmentation with suitable example.

Integrity Constraints

20. What are integrity constraints? Explain the various types of integrity constraints.

Client-Server Architectures (2-Tier vs 3-Tier)

21. Describe the following client-server architectures with the help of a diagram:

- (i) 2-tier (ii) 3-tier

File-Organization Techniques (Hashed & Inverted Files)

22. What is a hashed file organization and inverted file organisation? What are its advantages and disadvantages?

Transaction Control & Recovery

23. Explain the meaning of the following two keywords: Commit and Rollback. Write a code fragment for transferring money from account A to account B and show the uses of Commit and Rollback.

24. What are Checkpoints? How does this technique of Checkpoints contribute to database recovery? Give suitable example.

Data Dictionary

25. What is a data dictionary? What should be included in data dictionary?

Query Optimization & Relational Algebra

26. What is Optimization in DBMS? How does relational algebra help in Optimizing the Query? Discuss with suitable example. List the operators used in relational algebra and briefly explain them.

Shadow Paging vs. Log-Based Recovery

27. Explain log-based recovery scheme and shadow paging recovery scheme with the help of an example.